

Gursewak Singh

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Research Interests

Particle Physics, Dark Matter Physics, Cosmology, Computational Astrophysics, General Relativity, Quantum Field Theory

Education

M.Sc. Physics, SVKM's Mithibai College, Mumbai University 2024–2026

Overall CGPA: **7.76/10**

B.Sc. Physics, Chemistry, Mathematics, Kurukshetra University 2020–2024

Overall Marks: 2022/2900

Research Experience

Master's Thesis Student, Indian Institute of Technology Hyderabad Nov 2025 – Mar 2026

Simulation of Dark Matter Capture and Thermalization using Monte Carlo Methods

- Developed Python-based Monte Carlo simulations to model the capture and thermalization of Weakly Interacting Massive Particles (WIMPs) inside the Sun.
- Focused on spin-independent (Operator 1) WIMP–nucleon interactions within the non-relativistic effective field theory framework.
- Simulated particle trajectories, scattering events, and energy loss using numerical techniques and solar models.

Research Intern, Physical Research Laboratory (PRL), Ahmedabad May–Jun 2025

Destruction Mechanism of Presolar Grains

- Developed Python simulations to investigate destruction and survival of presolar silicate grains in supernova environments.
- Studied the effects of grain size, velocity and temperature using published astrophysical models.

Conference Presentation

Poster: Destruction of Silicate Grains

5th METMESS Conference, Indian Institute of Space Science and Technology (IIST), 2025

Workshops & Schools

- Gravitational Wave Open Data Workshop (2025)
- Winter School, National Astronomical Observatory of Japan (2025)
- Winter School, IISER Berhampur (2024)
- Data-Driven Astronomy (Coursera)

Technical Skills

Programming: Python (NumPy, SciPy, Matplotlib)

Scientific Tools: L^AT_EX, Mathematica, Excel

Version Control: Git

Operating Systems: Linux, Windows

Languages: Punjabi (Native), English, Hindi